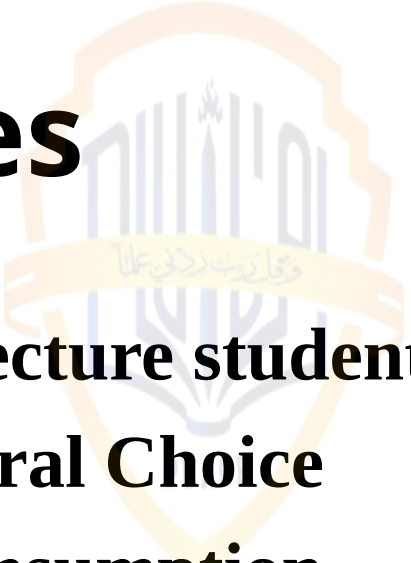




Intertemporal Choice

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Outlines

The logo of the University of Al-Qadisiyah is a circular emblem. It features a central shield with Arabic calligraphy. Above the shield is a banner with more calligraphy. The entire emblem is surrounded by a decorative border.

- **After this lecture students will be able to understand:**
- **Intertemporal Choice**
- **Present Consumption**
- **Future Consumption**
- **Factors influencing the intertemporal choice**
- **Time Preferences**
- **Discounting**
- **Opportunity Cost:**



- **Budget Line**

Interest rate and its different types

Intertemporal Choice

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- Intertemporal choice is the decision-making process involving the trade-off between present and future consumption, where individuals allocate resources considering time preferences, discounting, and opportunity costs.
- Intertemporal choice refers to the decision-making process involving the allocation of resources or consumption preferences over different periods of time.

Intertemporal Choice

- It encompasses individuals' decisions on how to balance present and future well-being, considering factors such as time preferences, discounting, and opportunity costs. In economics, it is a key concept for understanding how individuals, firms, and policymakers make choices that involve trade-offs between immediate gratification and future benefits

Explanation with Example:

- **Example: Saving for Retirement**
- Imagine a person named Asad, who earns a monthly salary. Asad faces an intertemporal choice when deciding how much of the salary to spend on immediate consumption (like dining out, entertainment, etc.) and how much to save for retirement.

Present vs Future Consumption

- **Present Consumption:**

- If Asad decides to spend a significant portion of the salary on present consumption, enjoying a luxurious lifestyle today, it provides immediate satisfaction and gratification. However, this choice may result in limited savings for the future.

- **Future Consumption (Saving for Retirement):**

- On the other hand, if Asad decides to allocate a portion of the salary to a retirement savings account, it represents an intertemporal choice favoring future well-being. By sacrificing some immediate consumption, Alex aims to ensure a financially secure and comfortable retirement.

Factors influencing the intertemporal choice

- **Time Preferences:** Individuals often have preferences for present versus future consumption. Some individuals may prioritize immediate gratification, while others may be more willing to delay consumption for greater future benefits.
- **Discounting:**
 - The perceived value of future benefits. If a person discounts future well-being significantly, the motivation to save for retirement may be lower.

Factors influencing the intertemporal choice

- **Opportunity Costs:**
- The concept of opportunity cost is essential in intertemporal choice. Individuals consider the potential benefits they forgo by choosing one consumption option over another, either in the present or the future.

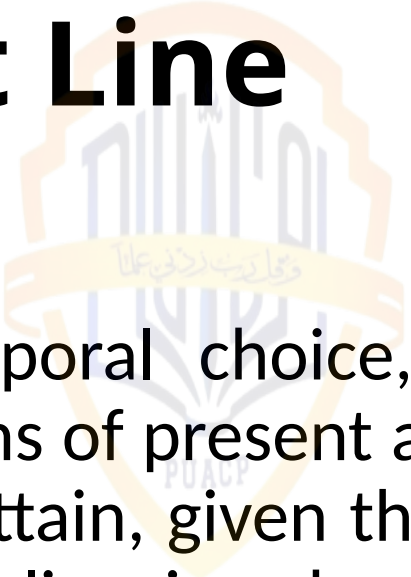
Factors influencing the intertemporal choice

- **Budget Constraints:** Individuals face constraints on their consumption choices based on their current and future incomes. The budget constraint in intertemporal choice illustrates the feasible combinations of present and future consumption given these constraints.

Factors influencing the intertemporal choice

- **Savings and Investment:** Intertemporal choices include decisions about saving money and investing in assets that yield future returns. Individuals weigh the benefits of saving for future consumption against the desire for immediate spending.

Budget Line

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- In intertemporal choice, the budget line represents the feasible combinations of present and future consumption that an individual or agent can attain, given their income and the prevailing interest rates. The budget line is a key concept in understanding how individuals allocate resources across different time periods.

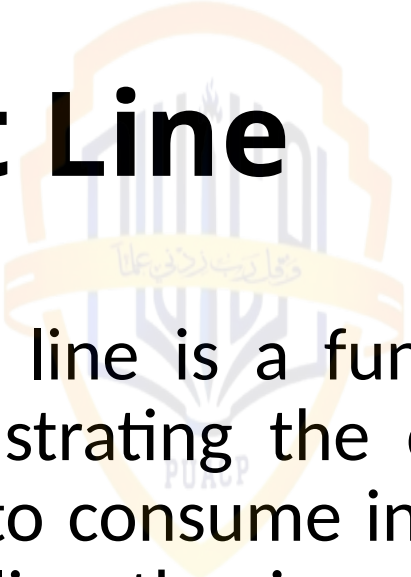
Budget Line

- The general form of the budget line in intertemporal choice can be expressed as:
- $C_1 + C_2/(1+r) = Y$
- Here:
- C_1 and C_2 are present and future consumption, respectively.
- Y is the individual's income.
- r is the interest rate.
- This equation states that the total expenditure on present and future consumption, discounted by the interest rate, should equal the individual's income

Budget Line

- The slope of the budget line is determined by the interest rate. If the interest rate is positive, the budget line has a negative slope, reflecting the trade-off between present and future consumption. As the interest rate increases, the opportunity cost of present consumption rises, encouraging individuals to save more and allocate resources toward future consumption.

Budget Line



- The budget line is a fundamental tool for analyzing intertemporal choices, illustrating the constraints individuals face when deciding how much to consume in the present versus saving for the future. It helps visualize the impact of interest rates on the intertemporal allocation of resources.

Interest Rate



- **Interest Rate**
- An interest rate is the cost of borrowing money or the return on investment expressed as a percentage of the principal amount. It plays a crucial role in various financial transactions, influencing borrowing, lending, and investment decisions. Different types of interest rates exist, each serving specific purposes. Here are some common types:

Different Types of Interest Rate

- **Nominal Interest Rate:**

- The nominal interest rate is the stated interest rate on a loan or investment without adjusting for inflation. It represents the actual interest charged or earned over a given period.

- **Real Interest Rate:**

- The real interest rate is the nominal interest rate adjusted for inflation. It reflects the actual purchasing power gained or lost on an investment or loan. The formula for the real interest rate is:

- $\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$
 $\text{Real Interest Rate} = \text{Nominal Interest Rate} - \text{Inflation Rate}$

Different Types of Interest Rate

- **Effective Interest Rate:**

- The effective interest rate, also known as the annual equivalent rate (AER) or annual percentage rate (APR), accounts for compounding. It represents the true cost or return on a financial product, factoring in how often interest is compounded within a year.

- **Simple Interest Rate:**

- Simple interest is calculated only on the principal amount over a specified period. It does not take into account any interest that accumulates on previously earned interest.

Different Types of Interest Rate

- **Compound Interest Rate:**

- Compound interest is calculated on the initial principal and the accumulated interest from previous periods. It reflects the compounding effect, where interest earned in one period contributes to the principal for subsequent periods.

- **Prime Rate:**

- The prime rate is the interest rate that banks offer to their most creditworthy customers. It serves as a benchmark for other interest rates in the economy, influencing rates for loans and credit cards.

Different Types of Interest Rate

- **Federal Funds Rate:**

- The federal funds rate is the interest rate at which banks lend to each other overnight in the federal funds market. It is a key tool used by central banks, like the U.S. Federal Reserve, to implement monetary policy.

- **LIBOR (London Interbank Offered Rate):**

- LIBOR is an interest rate at which major international banks lend to one another. It serves as a reference rate for many financial instruments worldwide.

Different Types of Interest Rate

- **Discount Rate:**

- The discount rate is the interest rate at which the Federal Reserve lends money to commercial banks. It is another tool used by central banks to influence monetary policy.
- Understanding the different types of interest rates is crucial for making informed financial decisions, whether borrowing, investing, or saving. Each type serves a specific purpose and may have different implications for individuals, businesses, and the overall economy.

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•Questions?



•**Thanks**